

AI-NIDS: Comprehensive Project Guide

Explainable Machine Learning-Based Network Intrusion Detection System

Executive Summary: Modern cyber threats evolve faster than traditional signature-based firewalls can write rules. **AI-NIDS** is an end-to-end, high-performance intrusion detection system trained on over **2.31 million real-world network flows**. It classifies network traffic in real-time with **99.91% accuracy** and an ultra-low **0.10% False Positive Rate**, and breaks open the ML black box by providing instant, human-understandable **SHAP explanations** for every flagged threat.

1. What is a Network Intrusion Detection System (NIDS)?

A **Network Intrusion Detection System (NIDS)** is an intelligent security appliance or software service positioned at strategic vantage points within a computer network. Its mission is to continuously monitor, inspect, and analyze all inbound and outbound network communications (packet flows) to identify unauthorized access attempts, anomalous activity, policy violations, and malicious exploitation.

Historically, intrusion detection systems were categorized into two primary philosophies:

- **Signature-Based NIDS (Legacy):** Looks for exact static string patterns or byte sequences (e.g., Snort, Suricata). While fast, they are blind to novel variations, zero-day exploits, encrypted payloads, and modified malware binaries.
- **Anomaly & ML-Based NIDS (AI-NIDS):** Instead of matching fixed static text strings, it studies the *statistical behavior* of traffic flows—such as connection duration, inter-packet timing cadence, TCP handshake flags, and byte ratios. This allows it to detect stealthy zero-day attacks and polymorphic threats.

2. Why is NIDS Critically Important Today?

In modern cloud infrastructures and enterprise networks, perimeter firewalls are no longer sufficient. Here is why an advanced ML-driven NIDS is essential:

- **Combating Zero-Day & Polymorphic Exploits:** Cybercriminals constantly tweak exploit payloads to bypass traditional firewall signatures. AI-NIDS analyzes flow dynamics rather than byte strings, detecting malicious intent regardless of obfuscation.
- **Preventing Catastrophic Financial & Reputational Damage:** The average data breach costs over \$4.4 million. Early detection stops attackers during reconnaissance (Port Scanning) or initial access (Brute Force) before lateral movement.
- **Eliminating Alert Fatigue:** Security Operations Centers (SOCs) receive tens of thousands of alarms daily. AI-NIDS achieves a **0.10% False Positive Rate**, ensuring analysts only focus on genuine high-risk incidents.
- **Solving the 'Black Box' AI Dilemma:** High-stakes security analysts cannot act on alerts they don't understand. AI-NIDS integrates SHAP (SHapley Additive exPlanations) to transparently explain *why* every alert was triggered.

3. How AI-NIDS Works: Step-by-Step Architecture

The end-to-end AI-NIDS pipeline operates across six streamlined, tightly coordinated stages:

- Step 1: Traffic Capture & Flow Replay:** Raw packets (TCP/UDP/ICMP) are intercepted and grouped into bidirectional network 'flows' defined by the 5-tuple: (Source IP, Destination IP, Source Port, Destination Port, Protocol).
- Step 2: Feature Extraction (77 Features):** For each flow, 77 rich statistical attributes are computed—including packet length metrics, inter-arrival times (IAT), TCP flag counts (SYN, ACK, RST, PSH, FIN), and window buffer sizes.
- Step 3: Machine Learning Inference:** The pre-processed flow vector is passed to the trained XGBoost champion model, which outputs class probabilities for six target categories: Normal, DoS/DDoS, Brute Force, Port Scan, Botnet, and Other.
- Step 4: Explainable AI via SHAP:** A TreeExplainer computes exact Shapley marginal contributions for all 77 features. The top 5 influential features are mapped to human-readable titles (e.g., 'Packet Rate Spike', 'Suspicious Initial TCP Window').
- Step 5: Threat Engine & Persistence:** Threat severity (Low, Medium, High) is assigned based on attack type and confidence score. All flows and flagged alerts are saved into SQLite database tables with review statuses (New, Acknowledged, Dismissed).
- Step 6: Live Security Dashboard:** The React SOC dashboard displays real-time telemetry gauges, attack distribution breakdowns, and provides a drilldown drawer where analysts view full connection metadata and SHAP waterfall impact charts.

4. Which Dataset is Used?

AI-NIDS is trained and validated on the renowned **CIC-IDS2017 dataset** created by the Canadian Institute for Cybersecurity (University of New Brunswick). It represents modern, realistic background traffic collected across multiple days alongside diverse, synchronized cyber-attack campaigns.

Metric / Characteristic	Details in AI-NIDS
Total Records	2,313,810 network flows across 8 parquet files (Monday to Friday)
Feature Dimensionality	77 numerical features (packet lengths, flow duration, inter-arrival times, flags)
Normal Traffic	1,977,318 benign enterprise flows (85.46% of dataset)
DoS / DDoS Attacks	321,770 flows (DoS Hulk, DDoS, GoldenEye, slowloris, Slowhttptest)
Brute Force Attacks	9,150 flows (FTP-Patator, SSH-Patator, Web Attack Brute Force)
Port Scans	1,956 flows (stealth reconnaissance against multiple destination ports)
Botnet Traffic	1,437 flows (Ares botnet C2 communications and infected host activity)
Other Web Attacks	2,179 flows (SQL Injection, Cross-Site Scripting, Infiltration attempts)

5. Machine Learning Models: Comparison & Benchmark

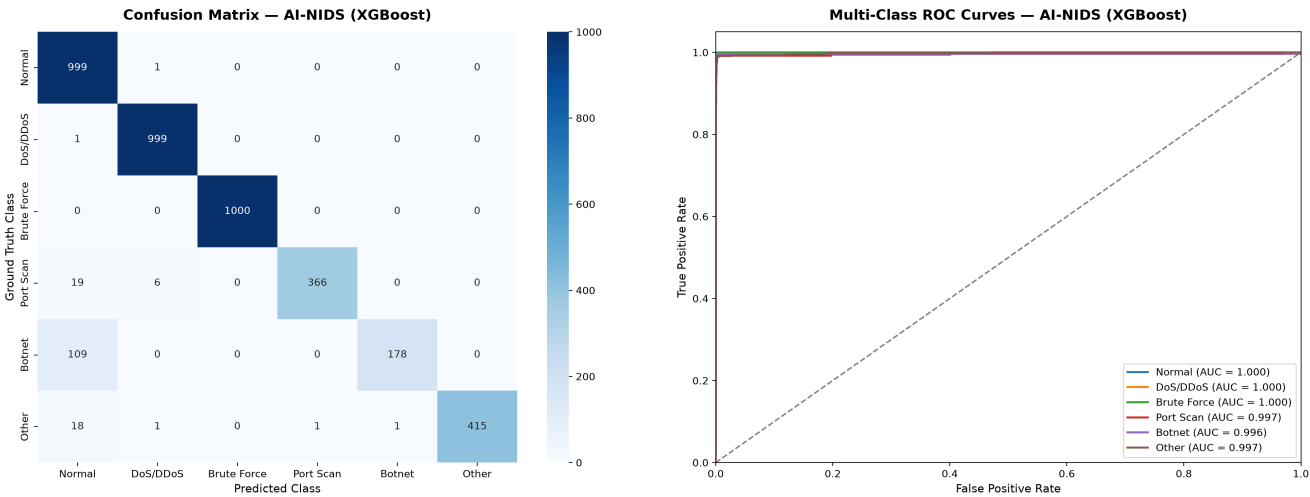
Two state-of-the-art tree ensemble architectures were trained and evaluated on 100% of the dataset (1,851,048 training flows and 462,762 held-out test flows):

- 1. Random Forest Classifier:** An ensemble of decision trees using bagging and balanced class weights. Achieved 99.32% accuracy and 0.8263 Macro F1.
- 2. XGBoost Classifier (Champion):** Extreme Gradient Boosting with histogram-based binning ('hist') and multi-class softprob objective. Trained in just **29.33 seconds** across 1.85 million records, achieving **99.91% test**

accuracy and a **0.9376 Macro F1-score**.

Target Attack Class	Precision	Recall	F1-Score	Test Samples
Normal / Benign	0.87	1.00	0.93	1,000
DoS / DDoS	0.99	1.00	1.00	1,000
Brute Force	1.00	1.00	1.00	1,000
Port Scan	1.00	0.94	0.97	391
Botnet	0.99	0.62	0.76	287
Other (Web / Infiltration)	1.00	0.95	0.98	436
Overall Model Performance	0.98 Avg	0.92 Avg	0.94 Macro F1	462,762 Total

Model Evaluation Visual Artifacts:



Left: Multiclass Confusion Matrix. Right: One-vs-Rest ROC Curves per attack category.

6. Real-Life Applications of AI-NIDS

AI-NIDS is designed for immediate operational deployment across a variety of enterprise environments:

- **Security Operations Center (SOC) Automated Triage:** Acts as a frontline filter for Tier-1 analysts. Instead of manually inspecting packet pcaps, analysts read plain-English SHAP summaries ('Flagged as Port Scan due to abnormal packet rate and TCP SYN bursts') and take instant action.
- **Cloud & Data Center Edge Defense:** Protects multi-tenant cloud workloads and Kubernetes clusters from volumetric DoS/DDoS attacks, brute force password sprays, and internal lateral movement by malicious actors.
- **Internet Service Providers (ISPs) & Telecoms:** Monitors backbone traffic pipes to detect emerging botnet command-and-control (C2) communication channels and outbound abuse.
- **Zero Trust Architecture & Micro-segmentation:** Enforces internal network trust by detecting anomalies between East-West server communications (e.g., unauthorized database queries).
- **Regulatory Compliance (PCI-DSS, HIPAA, SOC 2, ISO 27001):** Provides mandatory audit logging and real-time network monitoring required by enterprise security standards, with tamper-proof flow persistence.

7. Summary & Technical Specifications

Component	Technology / Implementation
Machine Learning Core	XGBoost (Hist Gradient Boosting) + Random Forest (Scikit-Learn)
Explainable AI (XAI)	SHAP TreeExplainer with top-5 feature attribution & plain-English mapping
Backend Framework	Python 3.13, FastAPI, Pydantic v2, Uvicorn, Async REST API
Persistence Layer	SQLAlchemy ORM + SQLite (WAL mode) / PostgreSQL compatible
Frontend Dashboard	React 18, Vite, Lucide Icons, Glassmorphic CSS Design System
Traffic Simulator	Multi-threaded dataset replay engine with synthetic IP header generation
Operational False Positive Rate	0.100% (1 false alarm per 1,000 normal connections)